UPSA
UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA
Scholarship with Professionalism

NETWORK AND SYSTEMS ADMINISTRATION

# WEEK 8: DATA CENTER OPERATIONS

Storage • Racks • Power/UPS • Cooling • Physical Security • Redundancy • Monitoring • Documentation

Practical Environment: Kali Linux VM

Dr Samuel Attuquayefio | Mr. Adams Azamati

# WEEK 8 ROADMAP

From Linux evidence to physical infrastructure decisions

| Inspect | Interpret | Design | Protect | Document |
|---|---|---|---|---|
| Storage, mounts, logs, memory and uptime | Capacity, health, risk and service impact | Power, cooling, racks, security and monitoring | Redundancy and single points of failure | Operational records and room layout |

## INDIVIDUAL PRACTICAL

Use lsblk -f, df -hT, findmnt, du -sh /var/log 2>/dev/null, free -h and uptime to inspect the Linux VM.

## GROUP ACTIVITY

Design a small server-room layout and identify power, cooling, storage, redundancy, physical security and monitoring requirements.

# WEEK 8 LEARNING OUTCOMES

By the end of the session, students should be able to:

- Explain the role of data-centre and server-room operations in IT service delivery.
- Identify storage, racks, power/UPS, cooling, physical security, redundancy, capacity, availability, environmental monitoring and documentation requirements.
- Use Linux commands to inspect storage capacity, mounted filesystems, disk usage, memory and uptime.
- Interpret command output as administrative evidence.
- Design a simple server-room layout for a Ghanaian organisational scenario.
- Identify at least three single points of failure and propose practical corrections.
- Explain why documentation supports maintenance, accountability and continuity.

# WHY DATA CENTER OPERATIONS MATTER

Services depend on the environment that houses, powers, cools, protects and monitors infrastructure.

- Users see applications, email, files, portals, Internet access and transaction systems.
- Behind the visible service are storage, power, cooling, racks, cables, access controls and monitoring.
- A data-centre fault often appears to users as an application or network fault.
- The administrator must keep the environment reliable, documented and ready for failure.
- Operational negligence can defeat good server configuration.

**ADMIN QUESTION**

What could stop the service even when the application code is correct?

**GHANA CONTEXT**

Consider a clinic, pharmacy, school, SME, bank branch or university department.

# DATA CENTER, SERVER ROOM OR EQUIPMENT CORNER?

The scale differs, but the operational principles are similar.

- A data centre is a purpose-built facility with controlled power, cooling, security, network, racks and monitoring.
- A server room is a smaller organisational space that hosts servers, storage, network equipment and UPS.
- An equipment corner is usually risky because critical devices are placed in ordinary office space.
- Small Ghanaian organisations still need controlled access, power planning, cooling and documentation.
- The course focus is practical administration, not expensive enterprise design.

**ADMIN QUESTION**

What could stop the service even when the application code is correct?

**GHANA CONTEXT**

Consider a clinic, pharmacy, school, SME, bank branch or university department.

# THE WEEK 8 OPERATIONAL SCOPE

Every outline item answers one question: can the infrastructure keep services available?

- Storage: capacity, filesystem location, reliability and backup implications.
- Racks: safe arrangement, labelling, access and airflow.
- Power/UPS: protection from interruption, surges and abrupt shutdown.
- Cooling: heat removal, airflow, temperature and humidity control.
- Physical security: authorised entry, tamper prevention and accountability.
- Redundancy, capacity and availability: what survives failure and growth.
- Environmental monitoring and documentation: how problems are detected and knowledge preserved.

**ADMIN QUESTION**

What could stop the service even when the application code is correct?

**GHANA CONTEXT**

Consider a clinic, pharmacy, school, SME, bank branch or university department.

# SERVICE AVAILABILITY CHAIN

Availability is the result of many working dependencies, not one server.

- User need depends on the application being reachable and usable.
- The application depends on the server or virtual machine.
- The server depends on storage, memory, power and cooling.
- The service also depends on network connectivity and secure access.
- Monitoring and documentation help administrators detect and restore failures.

**ADMIN QUESTION**

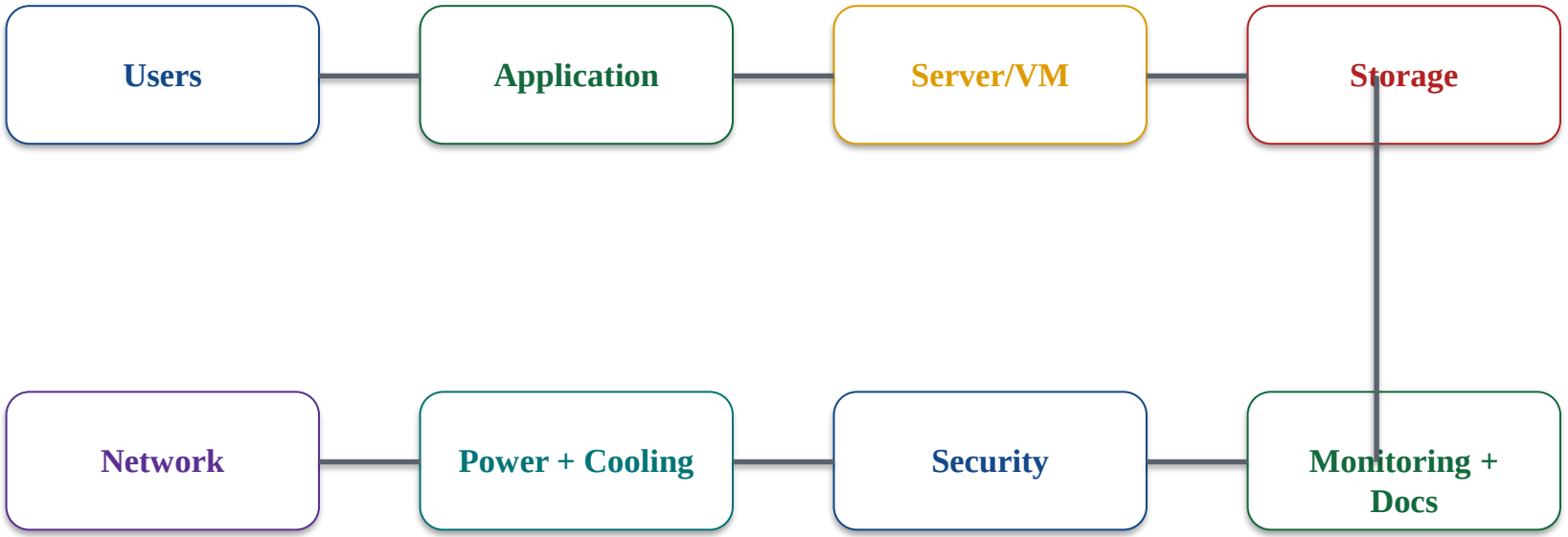
What could stop the service even when the application code is correct?

**GHANA CONTEXT**

Consider a clinic, pharmacy, school, SME, bank branch or university department.

# VISUALISING THE DEPENDENCY CHAIN

A weak link in operations can produce a visible service failure.




**Operations is the discipline of keeping these dependencies visible, monitored, protected and documented.**

# STORAGE CAPACITY AS ADMINISTRATIVE EVIDENCE

Storage evidence answers what exists, what is mounted and how much capacity remains.

- Capacity problems can stop services, break updates, corrupt logs and interrupt backups.
- Administrators inspect both block devices and mounted filesystems.
- A disk can exist without being mounted; a mounted filesystem can be full even when another disk has free space.
- The Week 8 commands create evidence for storage, mounts, disk usage, memory and uptime.
- Students must record the command, relevant output and interpretation.

**RUN / ASK**

Can this system continue to store, process and support services safely?

**RECORD**

Command • relevant output • what it means • any risk indicated

# COMMAND 1: lsblk -f

Inspect block devices, filesystems, labels, UUIDs and mount points.

- Shows disks and partitions in a device tree.
- The -f option adds filesystem information such as FSTYPE, LABEL, UUID and MOUNTPOINT.
- Useful for identifying whether a device has a filesystem and where it is mounted.
- Distinguishes the disk itself from partitions and mounted filesystems.
- Record which device appears to host the root filesystem.

**RUN / ASK**

$ lsblk -f

**RECORD**

Command • relevant output • what it means • any risk indicated

# COMMAND 2: df -hT

Inspect mounted filesystem usage in human-readable form.

- df reports space usage for mounted filesystems.
- -h gives human-readable sizes such as M, G or T.
- -T includes filesystem type for documentation and troubleshooting.
- Pay attention to Size, Used, Avail, Use% and Mounted on.
- High Use% on /, /var or /home may affect stability and service delivery.

**RUN / ASK**

$ df -hT

**RECORD**

Command • relevant output • what it means • any risk indicated

# COMMAND 3: findmnt

Inspect the current mount tree and filesystem relationships.

- Displays mounted filesystems as a hierarchy.
- Shows where filesystems are attached to the directory tree.
- Useful for confirming root, boot, removable media and special filesystems.
- Complements df because it emphasises relationships rather than only capacity.
- Record the most relevant lines if the output is long.

**RUN / ASK**

$ findmnt

**RECORD**

Command • relevant output • what it means • any risk indicated

# COMMAND 4: du -sh /var/log 2>/dev/null

Estimate disk space used by system logs.

- du estimates file and directory space usage.
- -s summarises the total rather than listing every file.
- -h displays human-readable size.
- /var/log matters because uncontrolled logs can fill a filesystem.
- 2>/dev/null hides permission messages so the output is cleaner for a beginner lab.

**RUN / ASK**

$ du -sh /var/log 2>/dev/null

**RECORD**

Command • relevant output • what it means • any risk indicated

# COMMAND 5: free -h

Inspect memory and swap availability.

- Shows total, used, free, shared, buff/cache and available memory.
- Linux uses memory for cache, so low free memory alone does not always mean failure.
- Available memory is often more useful for judging pressure.
- Swap information helps identify whether swap exists and whether it is being used.
- Connect memory evidence to capacity planning and service performance.

**RUN / ASK**

$ free -h

**RECORD**

Command • relevant output • what it means • any risk indicated

# COMMAND 6: uptime

Inspect how long the system has been running and current load average.

- Shows current time, system uptime, logged-in users and load averages.
- Uptime supports availability evidence but does not prove service health by itself.
- Load average should be interpreted with CPU capacity and workload context.
- A recent reboot may indicate maintenance, failure or patching.
- Record uptime as part of the Week 8 operational baseline.

**RUN / ASK**

$ uptime

**RECORD**

Command • relevant output • what it means • any risk indicated

# WEEK 8 PRACTICAL INVENTORY RECORD

Convert six command outputs into a capacity and operations record.

| Evidence item | Command | Administrative interpretation |
|---|---|---|
| **Block devices** | lsblk -f | Devices, filesystems, UUIDs, mount points |
| **Mounted capacity** | df -hT | Filesystem types and space pressure |
| **Mount tree** | findmnt | How filesystems attach to the directory tree |
| **Log usage** | du -sh /var/log | Whether logs are consuming notable space |
| **Memory** | free -h | Available memory and swap status |
| **Uptime/load** | uptime | Running time and load evidence |

**Evidence standard: command + output + interpretation + operational meaning.**

# PHYSICAL LAYOUT: CONTROLLED SPACE

A layout should make equipment stable, reachable, cool, protected and documented.

- Choose a location with controlled access, stable power and low exposure to water, dust and unauthorised traffic.
- Leave working space around racks and equipment for maintenance.
- Separate equipment from ordinary office activities and storage of unrelated items.
- Keep cabling neat, labelled and routed safely.
- Plan for airflow, UPS placement, fire safety and future expansion.
- Document the layout so another administrator can maintain it.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.

# RACKS AND EQUIPMENT ARRANGEMENT

Racks organise devices, cabling, airflow and maintenance access.

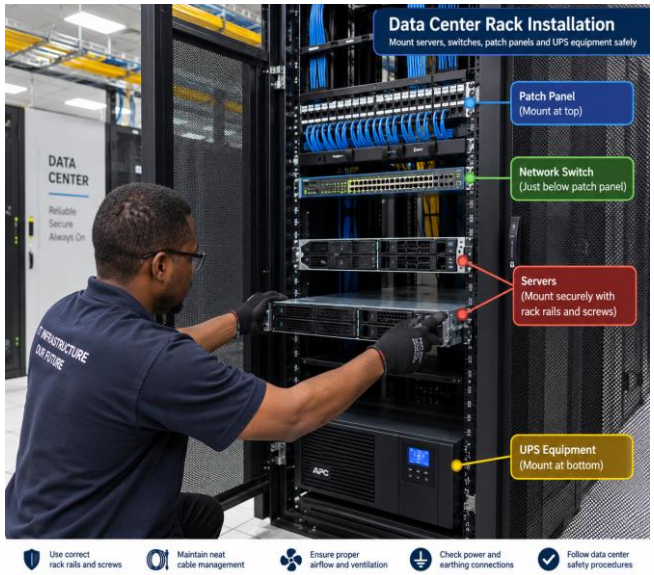
- Mount servers, switches, patch panels and UPS equipment safely.
- Keep heavier equipment lower to improve stability and safety.
- Label rack units, devices, ports and cables.
- Maintain front and rear access for service and airflow.
- Avoid cable clutter that blocks vents or makes troubleshooting difficult.
- Reserve space for growth rather than filling every position immediately.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.



# POWER: THE FIRST AVAILABILITY DEPENDENCY

No power means no service, regardless of good configuration.

- Power quality matters: fluctuation, outages and surges can damage equipment or cause shutdowns.
- Estimate how much power servers, switches, storage and cooling require.
- Avoid overloaded sockets and inappropriate extension boards.
- Document what is connected to which UPS, socket or circuit.
- Work with facilities or electricians, but understand the service risk.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.

# UPS: UNINTERRUPTIBLE POWER SUPPLY

A UPS gives controlled time to ride through outages or shut down safely.

- Protects equipment from abrupt power loss during short interruptions.
- Provides time for graceful shutdown when backup power is not available.
- May bridge the gap until a generator or alternative power source starts.
- Must be sized according to load and required runtime.
- Batteries age and must be tested, monitored and replaced.
- The UPS itself can become a single point of failure if poorly designed.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.

# COOLING: HEAT IS AN OPERATIONAL RISK

Computers work more reliably when heat is controlled.

- Servers, storage and network devices convert electrical power into heat.
- Excessive temperature can shorten equipment life or trigger failures.
- Cooling must match heat load from equipment, room design, lighting and people.
- Airflow matters: blocked vents and poor rack orientation reduce cooling effectiveness.
- Temperature and humidity should be monitored, not guessed.
- Cooling should allow for growth and seasonal conditions.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

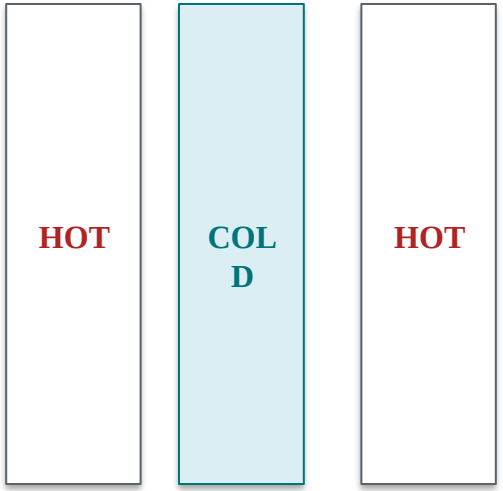
**GROUP LINK**

Include this item in your small server-room design.

# HOT AISLE / COLD AISLE CONCEPT

Arrange equipment so cool air enters the front and hot exhaust leaves the back.

- Equipment should draw cool air into the intake side.
- Hot exhaust should be directed away from the intake path.
- Rack orientation affects cooling efficiency.
- A room comfortable for people may not be optimised for computers.
- Temperature sensors should be placed where equipment actually receives air.



# PHYSICAL SECURITY

Cybersecurity fails when unauthorised people can touch critical equipment.

- Restrict server-room access to authorised personnel only.
- Use locks, access logs, visitor control and supervision where appropriate.
- Protect against theft, tampering, accidental unplugging and unauthorised console access.
- Keep backup media and administrative documentation secure.
- Avoid using the server room as unrelated storage.
- Balance emergency access with accountability.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.

# REDUNDANCY

Redundancy asks what keeps working when one component fails.

- Power redundancy may include UPS, generator, dual supplies or separate power paths.
- Network redundancy may include spare switches, alternate uplinks and configuration backups.
- Storage redundancy may include RAID, mirrored disks, NAS/SAN resilience and tested backups.
- Cooling redundancy may include standby cooling or alerts before overheating.
- Human redundancy means more than one person understands the environment.
- Redundancy must be tested and documented.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.

# CAPACITY MANAGEMENT

Capacity management prevents today's resources from becoming tomorrow's outage.

- Track storage growth, memory use, CPU load, network demand and environmental limits.
- Plan for users, services, data growth, backups and retention requirements.
- Use thresholds and trends rather than waiting for full disks or overheated rooms.
- Capacity includes room space, rack units, power, cooling and UPS runtime.
- Growth should be planned with budget, procurement and service priorities.
- Linux evidence from df, du, free and uptime can support capacity decisions.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.

# AVAILABILITY

Availability means services are usable when the organisation needs them.

- Different services require different availability levels.
- A clinic records system, payroll server and noticeboard may not need the same priority.
- Availability depends on application design, system administration and data-centre operations.
- Planned maintenance should be communicated and documented.
- Unplanned downtime should trigger incident recording and improvement action.
- Do not promise availability without understanding dependencies and failure points.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.

# ENVIRONMENTAL MONITORING

Monitor the room before the room silently damages the service.

- Monitor temperature at equipment intake and critical room locations.
- Monitor humidity where appropriate to avoid static or condensation risk.
- Monitor UPS status, battery health, load and runtime.
- Monitor storage thresholds and unusual log or data growth.
- Monitor links, reachability and critical services.
- Ensure alerts reach a responsible person who can act.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.

# OPERATIONAL DOCUMENTATION

Documentation turns a server room from a mystery into a manageable environment.

- Maintain a server-room layout and rack elevation diagram.
- Keep asset inventory with serial numbers, roles, owners and support details.
- Document power map: UPS, sockets, circuits and connected devices.
- Record cooling, environmental monitoring and maintenance events.
- Document network cabling, switch ports, patch panels and IP address records.
- Maintain incident, backup, change and escalation records.

**ADMIN QUESTION**

What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.

# SINGLE POINTS OF FAILURE

One component whose failure can stop the service.

- One UPS supporting all critical equipment with no alternative.
- One Internet router or switch connecting every user and server.
- One storage device holding production data and backup copies together.
- One air conditioner protecting the whole server room.
- One administrator who alone knows passwords, diagrams or procedures.
- One undocumented cable path that no one can trace during an outage.

**ADMIN QUESTION**

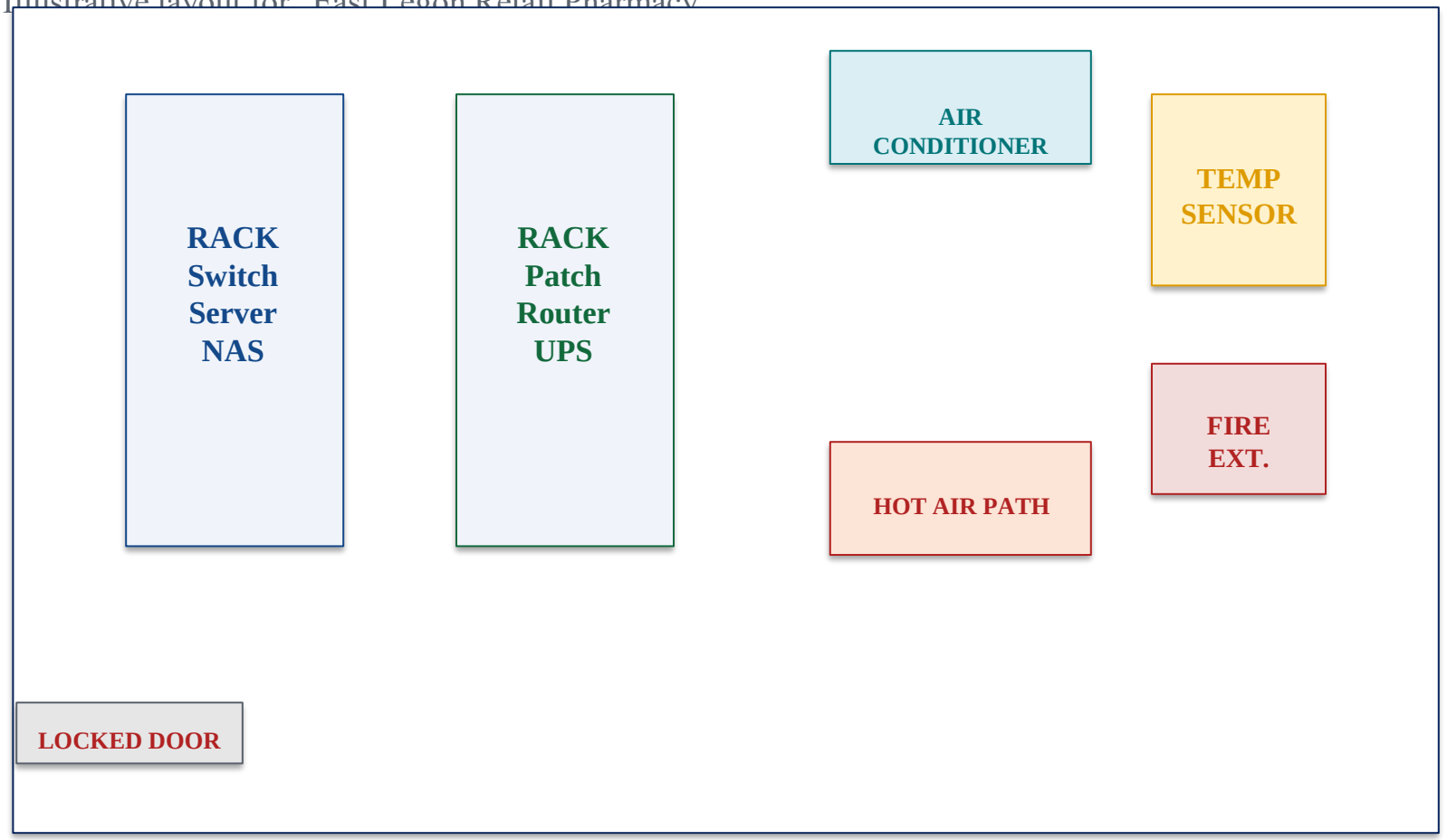
What evidence or control proves this item is being managed?

**GROUP LINK**

Include this item in your small server-room design.

# EXAMPLE: SMALL SERVER-ROOM DESIGN

Illustrative layout for "East Legon Retail Pharmacy"




**SPOFs to discuss: one AC unit • one router • one administrator with all passwords.**

# GROUP ACTIVITY: SERVER-ROOM DESIGN

Each group designs a small data-centre/server-room layout for its organisational scenario.

- Use the organisation selected earlier or choose a small Ghanaian organisation.
- Identify the services the room must support.
- Draw a simple room/rack layout showing major equipment.
- Identify power and UPS requirements.
- Identify cooling and airflow considerations.
- Identify storage and backup requirements.
- Identify physical security and monitoring controls.
- Identify at least three single points of failure and corrective actions.

**OUTPUT**

Diagram + responsibility notes + evidence/risk list.

**LECTURER CHECK**

Does the group cover every Week 8 outline item?

# GROUP WORKSHEET 1: SERVICES AND EQUIPMENT

Design begins with what the organisation must keep running.

- Organisation name and purpose.
- Critical services: records system, accounting, POS, email, website, learning system or file sharing.
- Users affected when the service fails.
- Servers, VMs, storage and network devices required.
- Backup devices or cloud backup link.
- Rack or shelf location for each device.
- Which device is most critical and why?

**OUTPUT**

Diagram + responsibility notes + evidence/risk list.

**LECTURER CHECK**

Does the group cover every Week 8 outline item?

# GROUP WORKSHEET 2: POWER, UPS AND COOLING

Plan what happens when power is unstable or heat rises.

- Which devices must connect to the UPS?
- How much runtime is needed for safe shutdown or generator transition?
- What device should not be connected because UPS capacity is limited?
- Where will cooling enter and where will hot air leave?
- Where should temperature sensors or alerts be placed?
- What happens if the air conditioner fails?
- What maintenance evidence should be recorded?

**OUTPUT**

Diagram + responsibility notes + evidence/risk list.

**LECTURER CHECK**

Does the group cover every Week 8 outline item?

# GROUP WORKSHEET 3: SECURITY, MONITORING AND DOCUMENTATION

A good layout must be operable by more than one trusted person.

- Who may enter the room, and how is access controlled?
- How are administrator passwords, keys and backup media protected?
- What environmental or system alerts are required?
- What must be labelled: devices, ports, cables, racks and sockets?
- What documents are needed: layout, inventory, power map, network map and maintenance log?
- Who receives alerts and who authorises changes?
- What three single points of failure have you identified?

**OUTPUT**

Diagram + responsibility notes + evidence/risk list.

**LECTURER CHECK**

Does the group cover every Week 8 outline item?

# GROUP PRESENTATION: 3 MINUTES EACH

Present as administrators responsible for availability and risk.

- 30 sec — Organisation and critical services.
- 30 sec — Major equipment and layout.
- 30 sec — Power/UPS and cooling plan.
- 30 sec — Storage, backup and monitoring plan.
- 30 sec — Physical security and documentation plan.
- 30 sec — Three single points of failure and corrective actions.

**OUTPUT**

Diagram + responsibility notes + evidence/risk list.

**LECTURER CHECK**

Does the group cover every Week 8 outline item?

# GOOD WEEK 8 EVIDENCE

Evidence connects commands, layout decisions and risk control.

## COMMAND EVIDENCE

Exact command, relevant output, interpretation and risk or capacity meaning.

## DESIGN EVIDENCE

Diagram, device roles, power/cooling/security/monitoring decisions and assumptions.

## RISK EVIDENCE

Single points of failure, likely impact and corrective action or monitoring plan.

**Strong administrative work is defensible: another person can see what you observed, why you decided, and what you documented.**

# THREE-HOUR SESSION FLOW

Balance lecture, command practice, group design and presentation.

- 0:00–0:35 — Lecture: data-centre/server-room operations and dependencies.
- 0:35–1:15 — Linux practical: lsblk, df, findmnt, du, free and uptime.
- 1:15–1:30 — Interpret command output and build evidence table.
- 1:30–2:10 — Group design activity: server-room layout and failure-point analysis.
- 2:10–2:50 — Group presentations and peer questioning.
- 2:50–3:00 — Key takeaways, evidence checklist and Week 9 preparation.

# COMMON BEGINNER MISTAKES

Avoid designs that look neat but fail operationally.

- Treating UPS as permanent power rather than short-term protection.
- Forgetting that cooling and airflow are part of availability.
- Putting production data and backup copies in the same failure zone without another copy.
- Drawing equipment without labelling roles, ports or dependencies.
- Assuming “locked room” means secure access without key control or logs.
- Ignoring cable management until troubleshooting becomes difficult.
- Submitting Linux output without interpretation.

# WEEK 8 KEY TAKEAWAYS & READING

Data-centre operations protect service availability by managing physical and operational dependencies.

- Storage, mounts, disk usage, memory and uptime can be inspected with Linux evidence.
- Racks, cabling and layout affect maintenance, airflow, safety and troubleshooting.
- Power, UPS and cooling are core availability controls, not facilities afterthoughts.
- Physical security protects systems from theft, tampering and accidental disruption.
- Redundancy and capacity planning reduce service risk but must be realistic and documented.
- Monitoring and documentation make the environment manageable beyond one person.
- Reading: UNIX and Linux System Administration Handbook, Chapter 30: Data Center Basics, pp. 1091–1104.